

Institutional-Grade Trading Rules for SPY Options Model Integration

Institutional-Grade Rulebook for SPY Options Trading in the Quantor-MTFuzz/CondorBrain System

Introduction

This report presents a comprehensive, institutional-grade rulebook for SPY options trading, specifically tailored for integration into the Quantor-MTFuzz and CondorBrain (Mamba-2) model creation process. The objective is to enhance entry and exit accuracy for iron condor strategies and related directional options trades (calls, puts, spreads, straddles) using a one-minute OHLCV time frame, multi-timeframe consensus, and a suite of precomputed, volatility-adaptive indicators. The rules are designed to be precise, interpretable, and directly compatible with both rule-based model building and generative machine learning pipelines, including fuzzy logic sizing and forward neural predictors.

This rulebook synthesizes the latest research, mathematical specifications, and engineering practices from the CondorBrain/Quantor-MTFuzz documentation and supporting technical reports. It is structured by strategy type (trend-following, mean reversion, breakout, momentum, volatility-based), with each rule including a name, description, logic conditions (with indicator thresholds), and entry/exit triggers. Where appropriate, tables summarize indicator thresholds and rule parameters. All rules are optimized for integration into the CondorBrain model training process and institutional-grade deployment.

System Architecture and Integration Points

1. CondorBrain/Quantor-MTFuzz System Overview

The CondorBrain system is built on a deep, 32-layer Mamba-2 Selective-Scan State Space Model (SSM) backbone, processing a high-fidelity, time-aligned dataset of 1-minute SPY OHLCV bars and options chain data (including Greeks, IV, IVR, and technical indicators). The architecture features:

- **Multi-Head Outputs:** Simultaneous prediction of iron condor policy parameters, regime classification, and multi-horizon price forecasts.
- **Fuzzy Logic Fusion Layer:** Aggregates neural and technical indicator signals for robust, interpretable decision-making.
- **Volatility-Adaptive Indicators:** Dynamic Bollinger Bands, Parabolic SAR, RSI, ATR, ADX, MACD, Stochastics, and volume, all modulated by geometric and topological features (e.g., curvature, persistent homology).
- **Multi-Timeframe Consensus:** Alignment of signals across 1m, 5m, and 15m timeframes for regime confirmation and noise reduction.
- **Risk Controls:** Portfolio-level Greeks tracking, drawdown caps, delta budgets, and spread management rules.
- **Composite Loss Functions:** Training objectives that combine Huber loss, Sharpe ratio maximization, drawdown minimization, and turnover penalties.
- **Production-Grade Engineering:** GPU-optimized training/inference (BF16, TF32, CUDA graphs), memory-efficient data handling, and deterministic, audit-ready execution flows [【scientific_spec.md†L1-L100】](#) [【Enhancing CondorBrain \(Mamba-2\) for Institutional-Grade SPY Options Trading.pdf†L1-L100】](#) [【Integrated Imp.pdf†L1-L100】](#) [【Condor Brain Mathematical Ref.pdf†L1-L100】](#) .

Integration Points:

- **Feature Pipeline:** All rules and indicators are computed as features in the institutional-grade input tensor, ensuring compatibility with both rule-based and generative ML models.
- **Fuzzy Logic Sizing:** Rules output interpretable signals (e.g., confidence scores, regime flags) that feed into the fuzzy logic position sizing engine (StateBinner, Q-table).

- **Forward Predictors:** Neural forecasts (e.g., Mamba-2, diffusion heads) provide regime-aware, probabilistic signals that can be used as constraints or features in the rule logic.
 - **Risk Management:** All entry/exit and sizing rules are subject to portfolio-level risk controls, including delta, gamma, vega, and drawdown limits.
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Dynamic, Volatility-Adaptive Indicators

2. Indicator Definitions and Computation

All indicators are computed in a volatility-adaptive, regime-aware fashion, leveraging geometric and topological features for enhanced robustness:

- **Dynamic Bollinger Bands:** Bandwidth and thresholds adapt to local volatility energy, as measured by curvature proxies and persistent homology signatures.
- **Parabolic SAR (PSAR):** ATR-scaled distance and adaptive acceleration factors, sensitive to regime shifts.
- **RSI (Relative Strength Index):** Weighted by volatility energy and geometric curvature, producing a dynamic RSI that responds to both trend and regime changes.
- **ATR (Average True Range):** Exponentially weighted, with regime-aware scaling.
- **ADX (Average Directional Index):** Normalized by volatility energy, providing a regime-invariant measure of trend strength.
- **MACD (Moving Average Convergence Divergence):** Volatility-normalized, with dynamic signal line adaptation.
- **Stochastics:** Adaptive windowing and scaling based on local volatility and regime features.
- **Volume:** Ratio to rolling mean, with regime-aware thresholds for liquidity confirmation.
- **Topological Features:** Persistent homology (e.g., Betti numbers, persistent entropy) and curvature proxies (e.g., kappa) provide regime signatures for

consolidation, breakout, and chaos detection 【Dynamic Market Indicators - Math and Pythonic integration for CondorIntelligence.pdf†L1-L100】 【scientific_spec.md†L1-L100】 .

Pythonic Implementation: All indicators are implemented as vectorized, auditable modules, with explicit analytic formulas and deterministic outputs. See `intelligence/indicators/manifold_volatility.py` and `tda_signature.py` for reference.

Multi-Timeframe Consensus and Feature Precomputation

3. Multi-Timeframe Consensus Logic

Multi-timeframe consensus is achieved by aligning indicator signals across 1m, 5m, and 15m bars. The consensus score is computed as:

$$\mu_{\text{MTF}} = 1 - |C_{\{1/5/15\}} - 0.5| \times 2$$

Where μ_{MTF} is the weighted consensus of directional signals (bullish, bearish, neutral) across timeframes. A consensus near 1.0 indicates strong agreement (ideal for iron condor entries), while values near 0.0 indicate trend or regime conflict (avoid entries).

Feature Precomputation: All indicators and consensus scores are precomputed and stored as features in the input tensor, ensuring efficient, lag-free access during both rule-based and ML-driven inference 【README.md†L1-L100】 【scientific_spec.md†L1-L100】 .

Rule Design for SPY Options Strategies

4. Rulebook Structure

The following sections present categorized, institutional-grade trading rules for SPY options, organized by strategy type. Each rule includes:

- **Name**

- **Description**
 - **Logic Conditions (with indicator thresholds)**
 - **Entry/Exit Triggers**
 - **Parameter Table (where applicable)**
 - **Integration Notes**
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A. Trend-Following Rules

Rule A1: Dynamic Bollinger Band Breakout with Volume Confirmation

Description: Identifies trend continuation using a volatility-adaptive Bollinger Band breakout and volume spike.

Logic Conditions:

- Dynamic Bollinger Bands squeeze (bandwidth below 10th percentile of rolling window).
- Price closes above upper dynamic band.
- Volume > 1.5x rolling 20-bar average.
- ADX (volatility-normalized) > 25 and rising.

Entry Trigger:

- Enter long (call spread or directional call) when price closes above upper band with volume confirmation and ADX > 25.

Exit Trigger:

- Exit when price closes below dynamic middle band or volume drops below average.

Parameter Table:

Indicator	Threshold/Condition
BB Squeeze	Bandwidth < 10th percentile
Volume	> 1.5 × 20-bar average
ADX	> 25 and rising

RSI (dynamic)	> 60 (optional confirmation)
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Integration Notes: All thresholds are volatility-adaptive; the rule can be encoded as a feature or constraint in the CondorBrain model. The dynamic BB and ADX are computed using geometric and topological features for regime invariance 【Dynamic Market Indicators - Math and Pythonic integration for CondorIntelligence.pdf†L1-L100】 【scientific_spec.md†L1-L100】 .

Rule A2: MACD Crossover with Dynamic Band Expansion

Description: Uses a volatility-adaptive MACD crossover in the direction of a dynamic Bollinger Band expansion.

Logic Conditions:

- Dynamic Bollinger Bands begin to widen (bandwidth increases > X% over last Y bars).
- MACD line crosses above signal line (for long) or below (for short).
- ADX (volatility-normalized) > 20.

Entry Trigger:

- Enter long when MACD crossover aligns with price breaking upper dynamic band.

Exit Trigger:

- Exit when MACD line crosses back or price closes below dynamic middle band.

Parameter Table:

Indicator	Threshold/Condition
BB Expansion	Bandwidth increase > 20% (Y=10)
MACD	Line crosses above signal (long)
ADX	> 20

Integration Notes: The MACD and BB expansion are computed using volatility-adaptive formulas; the rule is suitable for both direct execution and as a model feature.

Rule A3: ADX-RSI Trend Confirmation

Description: Confirms trend strength using ADX and times entries with RSI.

Logic Conditions:

- ADX (volatility-normalized) > 25 and rising.
- RSI (dynamic) > 70 (overbought) or < 30 (oversold), but in the direction of the trend.

Entry Trigger:

- Long: ADX > 25 and rising, RSI > 70 (momentum long).
- Short: ADX > 25 and rising, RSI < 30 (momentum short).

Exit Trigger:

- ADX peaks and begins to decline.
- RSI shows divergence or exits extreme zone.

Integration Notes: The dynamic RSI and ADX are computed as described in the indicator section; this rule is particularly effective in strong trend regimes.

B. Mean Reversion Rules

Rule B1: Dynamic Band Reversion with Fuzzy Confirmation

Description: Trades mean reversion at dynamic Bollinger Band extremes, confirmed by fuzzy logic consensus.

Logic Conditions:

- Price touches or breaks dynamic upper/lower band.
- Dynamic RSI > 70 (overbought) or < 30 (oversold).
- Fuzzy logic consensus score > 0.7 for mean reversion (i.e., MTF consensus, BB position, RSI, and volume all support reversion).

Entry Trigger:

- Enter short at upper band with bearish reversal candle and RSI > 70.

- Enter long at lower band with bullish reversal candle and RSI < 30.

Exit Trigger:

- Exit at dynamic middle band or when fuzzy consensus drops below 0.5.

Parameter Table:

Indicator	Threshold/Condition
BB Position	Price at or beyond band
RSI (dynamic)	> 70 (short), < 30 (long)
Fuzzy Consensus	> 0.7

Integration Notes: The fuzzy logic consensus is computed as a weighted sum of indicator memberships; this rule is directly compatible with the fuzzy sizing engine.

Rule B2: RSI Divergence with Dynamic Band Touch

Description: Combines RSI divergence with dynamic Bollinger Band touch for reversion.

Logic Conditions:

- Price makes new high/low touching outer dynamic band.
- RSI shows divergence (e.g., price makes higher high, RSI makes lower high).

Entry Trigger:

- Enter short on bearish divergence at upper band.
- Enter long on bullish divergence at lower band.

Exit Trigger:

- Exit at dynamic middle band or when RSI returns to neutral (40-60).
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C. Breakout Rules

Rule C1: Dynamic Band Squeeze with Breakout Candle

Description: Detects breakout from consolidation using a dynamic Bollinger Band squeeze.

Logic Conditions:

- Dynamic Bollinger Bands narrow significantly (squeeze condition: bandwidth < 10th percentile).
- Breakout candle closes outside upper or lower dynamic band.
- Volume > 1.5x rolling average.

Entry Trigger:

- Enter long on breakout above upper band with volume spike.
- Enter short on breakout below lower band with volume spike.

Exit Trigger:

- Exit when price closes back inside bands or hits target based on prior range.
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Rule C2: Persistent Homology Regime Shift

Description: Uses topological regime signature (persistent homology) to detect breakout or consolidation.

Logic Conditions:

- Persistent homology signature (e.g.,) drops sharply (indicating destruction of cycles, i.e., breakout).
- Dynamic consolidation score < 0.3.

Entry Trigger:

- Enter in direction of breakout when topological signature confirms regime shift.

Exit Trigger:

- Exit when persistent homology signature rises (return to consolidation).

Integration Notes: The persistent homology signature is computed using Takens embedding and rolling window persistent diagram analysis; see `tda_signature.py` for implementation.

D. Momentum Rules

Rule D1: RSI and Dynamic Band Confluence

Description: Uses RSI extremes with dynamic Bollinger Band touch for momentum trades.

Logic Conditions:

- Price touches upper dynamic band and $RSI > 70$ (momentum long).
- Price touches lower dynamic band and $RSI < 30$ (momentum short).

Entry Trigger:

- Enter long when price breaks upper band and $RSI > 70$.
- Enter short when price breaks lower band and $RSI < 30$.

Exit Trigger:

- Exit when RSI returns to neutral or price reverts to dynamic middle band.
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Rule D2: Volume Spike with Band Break

Description: Uses volume surge to confirm momentum breakout.

Logic Conditions:

- Price breaks upper/lower dynamic band.
- Volume $> 1.5x$ rolling average.

Entry Trigger:

- Enter in direction of breakout with volume confirmation.

Exit Trigger:

- Exit on volume drop or price reversal to dynamic middle band.
-

E. Volatility-Based Rules

Rule E1: Spread vs. High-Low Average Constraint

Description: Manages entries/exits for multi-leg options spreads based on spread width relative to average high-low range.

Logic Conditions:

- Compute average high-low (H-L) range over past N bars.
- If current spread width $\geq$ average H-L over N bars, do not close.
- If current spread width $<$ average H-L over N bars by threshold T, allow entry/exit.

Entry/Exit Trigger:

- Entry: Spread width $<$ (Average H-L over N bars - T).
- Exit: Spread width $\geq$ Average H-L over N bars.

Parameter Table:

Parameter	Description
N	Lookback period for H-L average
T	Threshold for spread width deviation
Spread Width	Difference between bid and ask or between legs of spread

Integration Notes: This rule is critical for managing iron condor and spread exits during volatility spikes; it is directly compatible with the fuzzy logic and neural sizing systems.

Rule E2: Dynamic Band Width Expansion

Description: Uses dynamic band width expansion to detect volatility regime shifts.

Logic Conditions:

- Dynamic Bollinger Band width increases $> X\%$ over last Y bars.

Entry Trigger:

- Enter in direction of breakout when width expansion confirmed.

Exit Trigger:

- Exit when width contracts or price reverts to dynamic SMA.
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Rule E3: Persistent Homology Chaos Detection

Description: Uses persistent homology to detect chaotic, high-volatility regimes.

Logic Conditions:

- Persistent homology signature (β_1) rises sharply, indicating emergence of new cycles (chaos).

Entry/Exit Trigger:

- Reduce position size or avoid new entries when chaos detected.

Integration Notes: This rule provides a regime-aware filter for risk management and is directly compatible with the fuzzy logic engine.

Fuzzy Logic Sizing and Policy Outputs

5. Fuzzy Logic Sizing System

The fuzzy logic sizing engine (StateBinner, Q-table) integrates all rule outputs and indicator features into a final position size and trade intent. The process is as follows:

1. **Feature Extraction:** All dynamic indicators, regime signatures, and neural forecasts are extracted as features.
2. **State Binning:** Continuous features are discretized into bins (e.g., volatility energy, consolidation score, breakout score, IVR).
3. **Q-Table Policy:** A tabular Q-learning policy maps binned states to action scores (e.g., buy_call, sell_call, buy_put, sell_put, hold).
4. **Fuzzy Aggregation:** Membership functions aggregate indicator signals into a confidence score.
5. **Defuzzification:** The final position size is computed as a weighted sum of action scores, normalized and scaled by risk constraints.

Example Policy Vector:

```
u_t = (Q(\text{buy_call}), Q(\text{sell_call}), Q(\text{buy_put}),  
Q(\text{sell_put}), Q(\text{hold}))
```

Integration: The fuzzy logic system is fully integrated with the CondorBrain neural outputs and all rule-based signals, ensuring interpretable, risk-aware sizing decisions 【Dynamic Market Indicators - Math and Pythonic integration for CondorIntelligence.pdf†L1-L100】 【scientific_spec.md†L1-L100】 .

Forward Predictors and Neural Forecast Integration

6. Neural Forecasting and Regime Detection

The CondorBrain system includes a forward predictor module (Mamba-2 neural network with optional diffusion head) that outputs:

- **Market Direction Probabilities:** Bullish, bearish, neutral.
- **Predicted Volatility Regime:** Low, medium, high.
- **Trade Intent and Confidence:** Actionable signal generation with confidence score.

These outputs are used as features and constraints in the rule logic, enabling regime-aware gating and dynamic adjustment of entry/exit thresholds. For example, if the neural predictor signals "high volatility regime," the system can automatically widen iron condor wings, reduce position size, or avoid new entries.

Integration: Neural forecasts are fused with technical and fuzzy logic signals at both the feature and decision levels, ensuring robust, adaptive trading behavior.

Risk Controls and Constraints for Options Structures

7. Portfolio and Trade-Level Risk Management

All rules and sizing decisions are subject to strict risk controls, including:

- **Spread Management:** The spread vs. H-L average rule (Rule E1) prevents premature exits or poor entries during volatility spikes.
 - **Delta Budgets:** Maximum net delta per trade and portfolio, enforced by the RiskManager.
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- **Drawdown Bounds:** Maximum daily and rolling drawdown limits, enforced at the portfolio level.
- **Turnover Penalties:** Excessive position changes are penalized in the composite loss function to reduce slippage and transaction costs.
- **Liquidity Gates:** Minimum volume and open interest thresholds for all option legs.
- **Slippage and Fill Modeling:** Realistic bid/ask and execution modeling in backtesting and live trading.

Integration: All risk controls are implemented as hard invariants in the CondorBrain execution spine, with fail-closed behavior on violation.

Composite Loss and Training Objectives

8. Composite Risk-Aligned Loss Function

The CondorBrain model is trained using a composite loss function that directly aligns with trading objectives:

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{pred}} - \lambda_2 \mathcal{L}_{\text{sharpe}} + \lambda_3 \mathcal{L}_{\text{dd}} + \lambda_4 \mathcal{L}_{\text{turn}}$$

Where:

- $\mathcal{L}_{\text{pred}}$: Predictive fidelity (Huber loss)
- $\mathcal{L}_{\text{sharpe}}$: Negative Sharpe ratio (maximization)
- $\mathcal{L}_{\text{dd}}$: Soft drawdown penalty
- $\mathcal{L}_{\text{turn}}$: Turnover penalty

Default Hyperparameters:

Interpretation: This loss function ensures that the model not only predicts accurately but also optimizes for risk-adjusted returns, drawdown minimization, and operational efficiency.

Volatility-Gated Attention, TopK MoE, and Neuro-Fuzzy Overlays

9. Advanced Model Enhancements

- **Volatility-Gated Attention:** Sparse attention layers are inserted at key points in the Mamba-2 backbone (e.g., layers 8, 16, 24), activating global context only during high-volatility regimes.
- **TopK Mixture-of-Experts (MoE):** Regime-routed expert networks specialize in different volatility states, with sparse routing for efficiency.
- **Neuro-Fuzzy Overlays:** Differentiable fuzzy logic layers constrain neural outputs using rule-based safety logic, ensuring institutional risk discipline.

Integration: These enhancements preserve scaling and sub-second inference, while providing regime specialization and interpretable safety controls.

Practical Engineering: GPU Training/Inference Optimizations

10. Production-Grade Engineering Practices

- **BF16/TF32 Mixed Precision:** Enables high-speed, high-precision training on A100/H100 GPUs.
- **CUDA Graphs:** Sub-second inference via static CUDA graph capture.
- **Gradient Accumulation and Checkpointing:** Memory-efficient training with large effective batch sizes.
- **Memory Fragmentation Control:** Environment variables (e.g., `PYTORCH_CUDA_ALLOC_CONF=expandable_segments:True`) prevent OOM errors.
- **LazySequenceDataset:** Zero-copy, memory-efficient data loading for 10M+ row datasets.

Integration: All engineering practices are fully documented and implemented in the CondorBrain training and inference pipelines.

Backtesting and Validation Protocols

11. Institutional-Grade Backtesting

- **Walk-Forward Validation:** Sequential train/test splits with rolling windows, mimicking real-world deployment.
- **Lag-Aware Alignment:** Strict timestamp alignment of spot and options data, with explicit lag policies and IV confidence decay.
- **Realistic Slippage and Fills:** Mark-to-market P&L, bid/ask modeling, and liquidity gates.
- **Portfolio-Level Metrics:** Net profit, max drawdown, Sharpe ratio, win rate, profit factor, expectancy ratio.
- **Regression Testing:** Golden datasets and deterministic replay for auditability.

Integration: All backtesting protocols are enforced in both research and production environments, ensuring robust, reproducible performance evaluation.

Example Rule Templates and Logic

12. Indicator-Based Example Rule

Rule Name: Dynamic Band Reversion with PSAR Flip

Description: Triggers a call entry when the price breaks the lower dynamic Bollinger Band five times, PSAR flips down, and price crosses above the dynamic BB median within three bars.

Logic Conditions:

- Price closes below lower dynamic BB for five consecutive bars.
- PSAR flips from uptrend to downtrend.
- Within the next three bars, price crosses above dynamic BB median.

Entry Trigger:

- Enter call (or call spread) on the bar where price crosses above median.

Exit Trigger:

- Exit when price closes above upper dynamic BB or PSAR flips up.

Parameter Table:

Condition	Value/Threshold
BB Lower Breaks	5 consecutive bars
PSAR Flip	Downtrend
Median Cross	Within 3 bars

Integration: This rule is directly compatible with both rule-based and generative ML pipelines, and can be encoded as a feature or constraint in the CondorBrain model.

Summary Table: Indicator Thresholds and Rule Parameters

Indicator	Thresholds / Conditions
Bollinger Bands	Dynamic, volatility-adaptive; squeeze = narrow bands
RSI	Overbought > 70, Oversold < 30, Neutral = 40-60
ADX	> 25 (trend), < 20 (range), normalized by volatility
MACD	Crossover confirmation with breakout direction
Volume	Spike = >1.5x 20-bar average
Spread	Compare to average H-L over N bars
PSAR	ATR-scaled, adaptive acceleration
Persistent Homology	Regime signature for consolidation/breakout/chaos

Conclusion

This rulebook provides a rigorous, institutional-grade framework for SPY options trading within the Quantor-MTFuzz/CondorBrain system. By integrating dynamic, volatility-adaptive indicators, multi-timeframe consensus, fuzzy logic sizing, neural forecasts, and robust risk controls, the rules enable precise, interpretable, and risk-aligned trading

decisions. All rules are engineered for seamless integration into both rule-based and generative ML pipelines, with full compatibility for forward predictors, fuzzy logic sizing, and institutional risk management.

The system's architecture, loss functions, and engineering practices ensure that trading performance is robust, reproducible, and aligned with real-world institutional standards. This rulebook serves as both a practical implementation guide and a theoretical foundation for ongoing research and deployment in advanced options trading systems.

References:

All mathematical definitions, engineering practices, and rule logic are directly sourced from the CondorBrain/Quantor-MTFuzz documentation and supporting technical reports, including:

- Loss Function Specification and Training Error Interpretation
- Dynamic Market Indicators - Math and Pythonic integration for CondorIntelligence
- Scientific Specification: CondorBrain & Mamba Architecture
- Enhancing the Mamba 2 Neural Network for Advanced Trading and ROI Optimization
- Enhancing CondorBrain (Mamba-2) for Institutional-Grade SPY Options Trading
- Integrated Implementation Strategy: Synthesizing CondorBrain Architectures
- Condor Brain Mathematical Reference Manual

All code snippets and formulas are implemented and validated in the production CondorBrain codebase. For detailed mathematical proofs and implementation details, see the referenced documents and Python modules.